Generic Singular-Reference Ref-GH Driver Implementation and Qualification Checkpoint

AthenaK Ref-GH development record

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Abstract

This report records the first implementation checkpoint for a generic puncture-adapted reference-frame generalized-harmonic system. A reference-independent first-order-state puncture-exponent estimator and a Gaussian-localized analytic singular reference were implemented and exercised locally. The analytic estimator and the prescribed reference-jet scan pass, but the required direct-finite-difference comparison fails for a scale invariance reason on the prescribed shell. Therefore the overall status is **stopped at the initial estimator hard gate**. A focused standard γ_2 slice also passes local algebra and bounded Minkowski checks. No closed-loop control, hyperbolic gauge-driver, stationary black-hole, SMR, or PVC qualification claim is made.

1 Provenance and scope

The mandated parent branch, its upstream-tracking reference, and the remote reference all equaled `0e248310a562c8a84327421eecf70f2f5d1da4a3` before editing. Work was performed on `codex/ref-gh-generic-singular-driver-20260825`; the Kokkos submodule is `6739bc623081648af9e752b616d9671527922cbf`. The local executable is a Release Kokkos Serial build with GCC 13.3.0. Existing work in other worktrees was not modified.

The three mechanisms remain conceptually separate:

Reference geometry	regularizes singular evolved components,
Hyperbolic gauge driver	controls the physical coordinate gauge,
γ_0/γ_2	controls GH and first-order constraints.

The reference/estimator slice and standard γ_2 additions are implemented in this checkpoint; the physical gauge driver is not.

2 Primary equations and convention map

Ordinary GH uses

$$C^a = \widehat{H}^a + g^{bc}\Gamma^a{}_{bc}, \tag{1}$$

while the background-covariant form is

$$C^a = H^a_{\mathrm{bg}} + g^{bc}\left(\Gamma^a{}_{bc} - \bar{\Gamma}^a{}_{bc}\right). \tag{2}$$

Equality of the physical constraint gives the sign relation

$$H^a_{\mathrm{bg}} = \widehat{H}^a + g^{bc}\bar{\Gamma}^a{}_{bc}. \tag{3}$$

The parent source currently specializes to the background wave-map choice, equivalently $\widehat{H}^a = -g^{bc}\bar{\Gamma}^a{}_{bc}$.

The runtime γ_2 additions and Einstein characteristic map now implement Eqs. (35)–(37) and (32)–(34) of Ref. [1]. The remaining implementation must add the conformal Gamma-driver with its auxiliary Υ^i from Eqs. (60)–(62) of Ref. [2]; and the improved $\widehat{H}_a, \theta_a$ driver from Eqs. (9) and (11), including the combined characteristic system in Appendix B, of Ref. [3].

3 Pointwise physical exponent estimator

At each actual Cartesian cell center, with $X^i = x^i - x^i_{\text{puncture}}$, the implemented production quantity is

$$q_{\text{loc}} = -\frac{1}{6} X^k \gamma^{ij} \partial_k \gamma_{ij}. \tag{4}$$

The physical metric and derivative are reconstructed from Ψ, Φ, and the reference two-jets; no additional derivative is taken. The offline lapse diagnostic is

$$p_{\text{loc}} = X^k \partial_k \ln \alpha, \qquad \alpha = (-g^{00})^{-1/2}. \tag{5}$$

The selection and weighting are

$$2h \le r < 8h, \qquad 8h < R_G/2, \qquad w_i = (2h/r_i)^3, \tag{6}$$

with weighted variance and $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. No clipping, rays, radial interpolation, binning, or puncture extrapolation is used.

For wormhole data,

$$\psi_W = 1 + \frac{M}{2r}, \qquad q_{\text{loc}}(r) = -2r \frac{d \ln \psi_W}{dr} = \frac{M}{r + M/2}. \tag{7}$$

The first-order-state calculation agrees pointwise to at most 1.6×10^{-15}. For the stationary trumpet it agrees with the independent profile derivative to at most 7.1×10^{-13}, and Minkowski is exactly zero.

The production estimate uses every valid shell point. For the independent FD comparison, a point is discarded if any fourth-order directional stencil spans a puncture coordinate, $|X^i| \le 2h$. The state and FD values below use the same puncture-clear subset.

h/M	W prod.	W safe	W FD	T prod.	T safe	T FD
1/16	1.32184	1.11591	1.11851	0.856373	0.791536	0.792402
1/24	1.48531	1.30752	1.31097	0.902691	0.856383	0.857347
1/32	1.58484	1.43073	1.43477	0.926983	0.891313	0.892331
1/48	1.70032	1.58002	1.58484	0.951826	0.927678	0.928754
1/64	1.76543	1.66723	1.67252	0.964342	0.946240	0.947345

Table 1: Weighted estimates on the required shell. The full production sample has 2,144 points and $N_{\text{eff}} = 595.547$; the safe comparison has 480 points and $N_{\text{eff}} = 370.885$.

4 Direct-FD hard-gate failure

The independent diagnostic reconstructs γ_{ij} on neighboring actual cell centers and applies AthenaK's fourth-order centered operator directly. The puncture exclusion substantially reduces the state–FD discrepancy, but it still grows from 0.00260102 to 0.00528934 for the wormhole and from 0.000865660 to 0.00110559 for the trumpet. The strict executable therefore exits nonzero.

This behavior follows from scale invariance. For $\gamma_{ij} = r^{-2q}\delta_{ij}$, write a selected cell as $x^i = h\xi^i$. Any fixed centered stencil obeys

$$D_h\gamma(h\xi) = h^{-2q-1}D_1\gamma(\xi). \tag{8}$$

Because $X^k = h\xi^k$ and $\gamma^{ij} = h^{2q}\delta^{ij}$, the discrete contraction $X^k\gamma^{ij}D_k\gamma_{ij}$ is independent of h at fixed r/h. Unless the stencil is exact for the singular power, the bias cannot converge away. The weighted pure-power limits for this stencil and shell are about 0.989337 for $q = 1$ and 1.575776 for $q = 2$.

Thus the claim "local puncture-exponent estimator established" is not made under the controlling same-shell comparison, despite the analytic success of the first-order-state path.

5 Standard γ_2 checkpoint

Runtime `gamma2` defaults to zero. For fixed $\gamma_1 = -1$, production now adds

$$\delta\Pi_{AB,t} = -\gamma_2\beta^I C_{IAB}, \qquad \delta\Phi_{IAB,t} = \alpha\gamma_2 C_{IAB}. \tag{9}$$

The device gate checks the forward/inverse characteristic map $u^{1\pm} = \Pi \pm s^I\Phi_I - \gamma_2\Psi$, the standard symmetrizer, and the frozen reduction and curl subsidiary damping to 3.33×10^{-16}. In a bounded robust-Minkowski CPU comparison at $t = 0.2$, the reduction norm changes by 0.9725 for $\gamma_2 = 0$ and by 0.7962 for $\gamma_2 = 1$; both runs remain finite. This is not long-time or GPU qualification.

6 Gaussian singular reference and prescribed trajectory

The new positive isotropic Cholesky factor is

$$\sigma_q(r,t) = \exp[-q(t)W(r)\ln(r/M)], \qquad W(r) = \exp[-(r/R_G)^2]. \tag{10}$$

A quintic endpoint-flat $q : 2 \to 1$ trajectory supplies analytic value, first derivative, and second derivative. Consequently the reference jets retain both

$$\partial_t \ln\sigma_q = -\dot{q}W\ln(r/M) \tag{11}$$

and the $\ddot{q}W\ln(r/M)$ and $\dot{q}^2W^2\ln^2(r/M)$ second-time-derivative structures.

The scan covers $\tau/M = 4, 8, 16$, $R_G/M = 2, 3, 4$, and $h/M = 1/16, 1/24, 1/32, 1/48, 1/64, 1/128$. Each dynamic midpoint has a static $q = 1.5$ control. The staged production cache agrees with the full geometry oracle to 1.11×10^{-15} at $t = \tau/2$.

Measure, $\tau = 8M, R_G = 3M$	$h = M/16$	$h = M/128$	model
$\lvert\dot{q}W\ln\rho\rvert$	0.683316	1.170901	log
$\dot{q}^2W^2\ln^2\rho$	0.466920	1.371010	log-squared
reference Ricci maximum	1.42809	4.11641	log-squared
total source maximum	2.85617	8.23283	log-squared

Table 2: Representative prescribed reference-jet scaling.

The compact analyzer finds no dynamic measure classified as unexpected positive algebraic growth. The prescribed-$q(t)$ reference-jet gate therefore passes locally. This result says nothing about physical evolution or feedback.

7 Qualification matrix and stop condition

Gate	Status
First-order-state estimator, analytic initial geometries	Passed locally
Direct-FD same-shell convergence	**Failed**
Prescribed Gaussian two-jets and staged cache	Passed locally on Kokkos Serial
Runtime γ_2 algebra and bounded robust Minkowski	Passed locally
Closed-loop q control	Not started; blocked by estimator gate
Dynamic $\widehat{H}/\theta/\Upsilon$ gauge system	Not implemented
Stationary/generic trumpet and wormhole evolution	Not run
SMR, restart, CUDA, Aurora PVC	Not run

No Aurora allocation was requested because the discriminator is local and mathematical. Before feedback work, the specification must choose between a fixed-physical-annulus direct-FD convergence check or a fixed-r/h known-bias comparison. Both avoid interpolation; neither is the literal present demand.

References

[1] L. Lindblom, M. A. Scheel, L. E. Kidder, R. Owen, and O. Rinne, "A New Generalized Harmonic Evolution System," arXiv:gr-qc/0512093.

[2] L. Lindblom, K. D. Matthews, O. Rinne, and M. A. Scheel, "Gauge Drivers for the Generalized Harmonic Einstein Equations," arXiv:0711.2084.

[3] L. Lindblom and B. Szilagyi, "An Improved Gauge Driver for the Generalized Harmonic Einstein System," arXiv:0904.4873.